Vigicaen: A `vigibase®` Pharmacovigilance Database Toolbox.

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Summary

Advanced methodologies are essential when conducting disproportionality analyses using pharmacovigilance data, as traditional approaches are susceptible to various biases such as reporting bias and confounding. The aim of vigicaen is to provide a toolbox for the VigiBase® Extract Case Level database, resolving technical challenges related to the large size of the database, and providing easier and reproducible access to advanced features. The package is built on top of the parquet file format. Functions related to drug and adverse event identification, and descriptive features such as time to onset, dechallenge, and rechallenge outcomes are provided. Command-line side-effect outputs aim at fast resolution of common issues related to drug and adverse event identification. The package is intended for pharmacovigilance practitioners, clinicians, and researchers with or without advanced biostatistical skills. A graphical output can be produced for routine use to support the daily assessment of drug liability.

Statement of need

Disproportionality analysis represents an essential component in the domain of drug safety signal detection. Advanced methodologies are required to address common biases within pharmacovigilance databases. These analyses necessitate expertise in biostatistical software, such as R, which may present substantial challenges in terms of acquiring and maintaining the requisite skills — in addition to a solid understanding of pharmacovigilance principles and reporting systems.

For decades, the World Health Organization (WHO) has been collecting adverse drug reaction reports, called Individual Case Safety Reports (ICSRs), from its member countries, populating more than 40 million reports to date. This pharmacovigilance database is called VigiBase® and is managed by the Uppsala Monitoring Centre in Sweden.(Uppsala Monitoring Centre, 2026) These ICSRs describe the course of patients who experienced an adverse event (a medical condition) after taking a drug. The burning question is whether this adverse event was actually related to the drug intake, e.g., if it is an adverse drug *reaction* (ADR). The pharmacovigilance database aims at uncovering the very first potential signals of association between drugs and ADRs.(Montastruc et al., 2011)

It relies on disproportionality analysis, a statistical method that produces estimators of how unlikely the number of observed ICSRs reporting on a specific drug and adverse event is to be attributable to chance alone. Together with an uncertainty margin, these estimators are used to raise safety signals on drugs.(Montastruc et al., 2011)

The Uppsala Monitoring Centre grants access to VigiBase® to researchers, either academic or industrial, under a license contract. The most extensive available version is called Extract Case

Level: it contains all the ICSRs, with information such as patient demographics, drug intake, adverse events, outcomes, dechallenge and rechallenge outcomes, and time to onset. However, this version is provided as large text files and requires a lot of processing before being usable for analysis. Those text files might be particularly challenging to use in R, as they would often exceed the size of the available Random Access Memory, thus requiring advanced knowledge of R computing techniques. Clinicians and pharmacovigilance practitioners typically lack these skills and therefore struggle to use VigiBase® data for their research. As a result, they often rely on partial data with limited statistical modeling options. Alternatively, they might develop home-made biostatistical scripts that are typically used once, often left undocumented, and highly heterogeneous across research teams.

The vigicaen package aims at providing a toolbox for the VigiBase® Extract Case Level database, tackling several technical challenges to run on low-specification computers, and providing easy and reproducible access to advanced features.(Dolladille & Chrétien, 2025) This article will explain the technical choices and data management logic underlying the package, and provide some examples of its main features. Additional examples and use cases are covered in the package vignettes, which can be found on the package website at https://pharmacologie-caen.github.io/vigicaen/. Of important note, the package is not supported by, nor does it reflect the opinion of, the WHO. The Uppsala Monitoring Centre, in charge of maintaining VigiBase®, was informed of the package development and kindly allowed its publication, acknowledging the potential benefit of promoting the use of VigiBase®.

State of the field

There are very few packages related to pharmacovigilance in R. Most are focused on interfacing with the Food and Drug Administration Adverse Event Reporting System (FAERS), and most attempt to visualize existing data, run basic disproportionality analyses, or perform web scraping. (Mukhopadhyay & Markatou, 2026) None of them actually allow browsing the entirety of a worldwide database such as VigiBase®, including all types of drugs and adverse events. Also, there is no existing package in the open-source community that prepares pharmacovigilance data in order to build advanced disproportionality metrics, such as machine or deep learning models. Finally, only a few of these packages are available on mainstream platforms such as CRAN.(Embry, 2016, 2018)

Research impact and significance

Our team and collaborators have already published several pharmacovigilance studies using vigicaen.Minoc et al. (2025). Vigicaen streamlines the data management process of pharmacovigilance studies, allowing for easier collaboration across centers around the world. The potential gain has already convinced several academic centers. The French Network of Regional Pharmacovigilance Centres is on its way to implementing vigicaen as part of routine practice across the 31 Pharmacovigilance Centres in France. The University of Nagoya has functional routines relying on vigicaen for disproportionality analyses. Vigicaen does not compete with existing open-source tools, but rather addresses an unmet need.

Software design

Key concepts were fundamental to building vigicaen: First, it should be open source, built on top of state-of-the-art practices to deal with large datasets (e.g., arrow), especially on low-specification computers, using a widespread and consistent syntax R users are familiar with (e.g., tidyverse). Although other syntaxes like `data.table` were once at the core of the package, they have now been phased out, as they were thought less fit for the project when considering the balance between performance and the user-facing interface. Second, it should

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address the most technically challenging issues for beginners in R or biostatistical software in general. Third, it should maintain as much rigor and consistency as possible in function naming, expected input formats, and outputs. Fourth, it should provide help, e.g., messages to users, to allow external checking of what is produced by the package. Fifth, it is not intended to implement model functions (like `glm`) per se, but rather to prepare the dataset so as to let the user run any model of their choice. Simple computations are nevertheless within scope (like bivariate disproportionality analysis or basic interaction analysis).

Open-source software practice

The package was developed according to best practices as promoted by R Packages, 2nd edition. Wickham & Bryan (2023) It is accompanied by a comprehensive set of unit tests (covering 100% of the code), in-depth documentation for each function and object, and several tutorial vignettes for both newcomers and advanced users. The source code is available on GitHub.com, so as to provide a unified platform to submit issues and propose pull requests. It is made available under the open-source CeCILL 2.1 license.

Development history

The first iteration of the package was built in 2020 as local software designed for internal use at Caen University Hospital. Later, it was called pharmacocaen and posted as a private repository on GitHub in 2022, due to intellectual property concerns with the Uppsala Monitoring Centre. After resolution of these property concerns, the package became available as a public repository on GitHub under the name vigicaen in 2024, and was accepted on CRAN in 2025. In the first versions, the package was mainly focused on performing vectorized data management so as to identify a large number of drugs and reactions in a compiled way. As there was a wide variety of settings under which drugs and reactions could be identified, bug fixing and handling edge cases were the main concerns for several years. Then, additional functionalities like building datasets from text files and descriptive statistics were added. Contacts were made with members of the Uppsala Monitoring Centre regarding their own work on other topics. These exchanges helped define the exact perimeter of vigicaen, as well as its potential articulation with other open-source software in the future. Also, vigicaen was discussed with end-users from pharmacovigilance centers in France, which led to the development of specific functions like `vigi_routine`.

Processing `vigibase®` source files.

Clinicians and pharmacovigilance researchers are used to working with low-specification computers. The typical available Random Access Memory rarely exceeds 16GB, which is one of the key bottlenecks when dealing with large data files in R.(Wickham et al., 2023) VigiBase® Extract Case Level files currently exceed 30GB once unpacked, which is too large to be loaded in-memory by mainstream readers like `read.table()`.

Vigicaen relies on `parquet` files, a recent format based on open standards.(Apache Software Foundation, 2026b) Arrow is a cross-language development platform that allows for the manipulation of large datasets.(Apache Software Foundation, 2026a) It is implemented in R via the arrow package.(Richardson et al., 2026) Datasets remain out of memory, allowing for the processing of large files on low-specification computers. Various tests of vigicaen on 16GB RAM computers succeeded in processing the source files. This, in combination with an alignment as close as possible to the tidyverse style guide, is also aimed at providing a modern and more rigorous approach compared to base R.(Wickham & RStudio, 2023)

Sourcing VigiBase® Extract Case Level files is done with the `tb_*` family of functions.

First, we define paths to the source folders.

```r
library(vigicaen)

path_base <- paste0(tempdir(), "/main/")
path_sub  <- paste0(tempdir(), "/sub/")

dir.create(path_base)
dir.create(path_sub)
```

Example files can be put in these folders.

```r
create_ex_main_txt(path_base)
create_ex_sub_txt(path_sub)
```

Then, we run the related `tb_*` function, `tb_vigibase()`.

```r
tb_vigibase(path_base, path_sub)
```

```
## -- tb_vigibase() -----------------------------------------------------------------
##
## i Checking for existing tables.
##
## i Creating vigibase tables.
##
## This process must only be done once per database version.
## It can take up to 30minutes.
## ==========>--------------------   33% | 1s | Remove duplicates
## ==========>--------------------   34% | 1.1s | Write srce.parquet
##
```

With an average computer, the real running time is around 20–30 minutes on the current database version.

If the dictionaries for drugs and adverse events are also required, `tb_who()` and `tb_meddra()` can be used.

Identifying drugs and adverse events

Exposure to drugs and occurrence of adverse events are located in the `drug` and `adr` tables, respectively. They connect together through the `demo` table, in a many-to-one relationship, via the `UMCReportId` key variable. Drugs and adverse events themselves are identified by codes (or IDs) from the WHO Drug Dictionary and the Medical Dictionary for Regulatory Activities (MedDRA), respectively. Disproportionality analysis requires a dataset with one row per ICSR, with the corresponding drugs and adverse events.

The following logic is implemented in vigicaen:

1. Use drug and adverse event names to collect their IDs.
2. Match the IDs in the `drug` and `adr` tables to identify the cases.
3. Report this information in `demo` (or any other VigiBase® table).

This is done with the `get_*` functions (step 1) and the `add_*` functions (steps 2 and 3). The overall process requires the sequential use of both. Below is an example to identify the drugs. The same principle applies to adverse events.

```r
# Load vigibase tables and drug dictionary
demo <- dt_parquet(path_base, "demo")
drug <- dt_parquet(path_base, "drug")
```

```r
# Here, example datasets
demo <- demo_
drug <- drug_
mp   <- mp_

# Select drug names
d_sel <- list(ipilimumab = "ipilimumab")

# Get the drug IDs
d_drecno <-
  get_drecno(
    d_sel,
    mp = mp
  )

# Add it to demo
demo <-
  demo |>
  add_drug(
    d_drecno,
    drug_data = drug
  )
```

```
## -- get_drecno() ----------------------------------------------------------------
##
## -- `d_sel`: Matching drugs --
##
## -- v Matched drugs
##
## > `ipilimumab`: "ipilimumab" and "ipilimumab;nivolumab"
## i Set `verbose` to FALSE to suppress this section.
##
## --------------------------------------------------------------------------------
```


Displaying information at the command line

As seen in the output above, the get_* functions do two things: they return drug or adverse event IDs (stored in d_drecno in the example), and they display command-line information about the matching process. This is especially useful since drug and adverse event names may vary in their spelling and case, while the underlying dictionary only accepts exact matches. Matched and unmatched names are displayed, along with some hints for the reasons for non-matching.

```r
meddra <- meddra_

a_sel <-
  list(colitis_term = c("Colitis", "Autoimmune colitis"),
       pneumonitis_term = "pneumonitis")

a_llt <- get_llt_soc(a_sel, term_level = "pt", meddra = meddra)
```

```
## -- get_llt_soc() ---------------------------------------------------------------
##
## -- v Matched reactions at `pt` level (number of codes) --
##
## > `colitis_term`: "Autoimmune colitis (1)" and "Colitis (25)"
```

```
## > `pneumonitis_term`: x No match
## i Set `verbose` to FALSE to suppress this section.
##
## -- x Unmatched reactions --
##
## -- ! Some reactions did not start with a Capital letter
##
## * In `pneumonitis_term`: x "pneumonitis"
```

The named list for inputting drug and adverse event names

The `get_*` and `add_*` functions are built with a named list as their first argument. This structure may seem a bit complex, especially for newcomers, but it allows for genuine flexibility when analysis plans increment. As an example, one may create `list(drug_group_1 = c("ipilimumab", "nivolumab"))` to automatically gather all ICSRs reporting one of these two drugs through `get_drecno()` and `add_drug()`.

Descriptive features

Descriptive features often play an important role in pharmacovigilance studies. They can be as important as producing statistical estimates to assess the liability of a given drug. Among them, the time to onset is rather challenging to compute. The main reasons are the uncertainty around the exact reported time to onset, and the potential for multiple reports for a given drug-adverse event pair in a single ICSR. The first is tackled by the Uppsala Monitoring Centre, which recommends in internal documentation to analyze ICSRs where the uncertainty interval is no more than a day. The second is addressed in `extract_tto()` or `desc_tto()`, which only extracts the longest time to onset reported for a given drug-adverse event pair in a given ICSR. This variable is called `tto_max`. Admittedly, this is a simplification that might not cover all potential use cases, for example, if the question is the time since the last infusion of a drug.

A similar simplifying approach is applied to drug dechallenge (`desc_dch()`) and rechallenge (`desc_rch()`) outcomes, as well as adverse event outcomes (`desc_outcome()`).

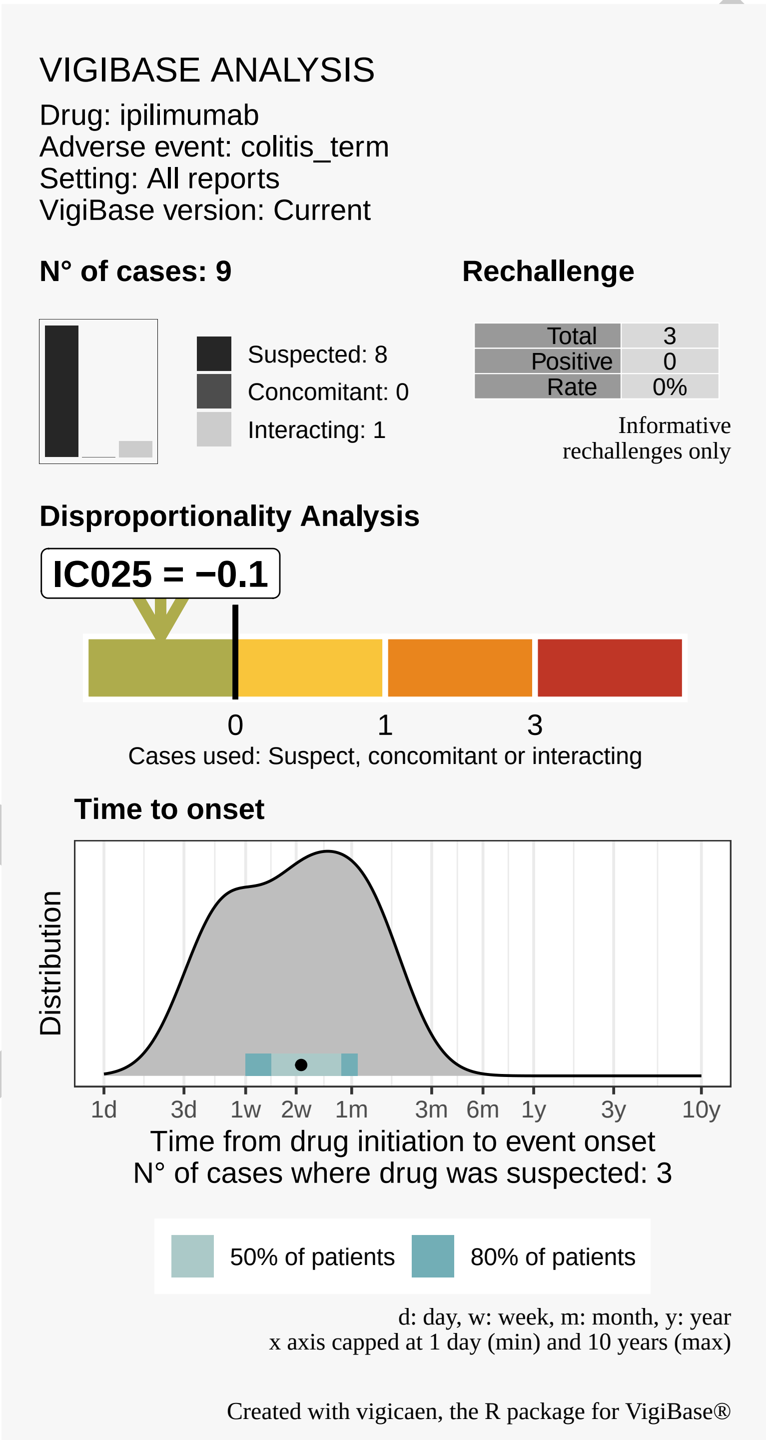
Disproportionality estimates

Generating disproportionality estimates, metrics aiming at evaluating the association between drug intake and adverse drug reaction occurrence, remains a cornerstone of safety signal detection in the field of pharmacovigilance.(Montastruc et al., 2011) Although the aim of the package is to prepare readily available datasets for users to compute disproportionality estimates on their own via advanced modeling techniques, it also provides basic estimates through the `compute_dispro()` and `compute_interaction()` functions. The underlying computations rely on the Norén et al. methodology for both point estimates, confidence, and credibility intervals.(Norén et al., 2013)

Routine use

For a routine pharmacovigilance practitioner, key information on a drug-adverse event pair may be needed out-of-the-box, without further need for manipulating the underlying tables. To address typical needs (disproportionality estimates, time to onset, dechallenge, and rechallenge outcomes), `vigi_routine()` creates a graphical output for a given pair. It is intended as a daily practice tool to support routine assessment of causality. The graph can easily be exported to an external file with the `export_to` argument.

```r
vigi_routine(
  demo,
  drug,
  adr_,
  link_,
  d_code = d_drecno,
  a_code = a_llt[1],
  vigibase_version = "Current"
)
```



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AI usage disclosure

GitHub Copilot and other AI assistants were episodically used during the software development. The main goals were to draft pull requests from existing issues, to assist with code syntax, and to improve the English writing in the documentation. It was especially useful for drafting variants of existing checkers or avoiding typographical errors when transforming larger sections due to architectural changes. It is foreseeable that such tools will play a growing role in the development of vigicaen, as is the case in many other settings. All AI-written code was human-checked by one of the package authors before being accepted. Generative AI was used to check spelling and improve the syntax of this manuscript.

Conclusion

Easier, reproducible research in pharmacovigilance databases is key to appropriate safety signal detection. Vigicaen proposes a set of tools based on popular open standards to facilitate pharmacovigilance analysis in R.

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References

Alexandre, J., Salem, J.-E., Moslehi, J., Sassier, M., Ropert, C., Cautela, J., Thuny, F., Ederhy, S., Cohen, A., Damaj, G., Vilque, J.-P., Plane, A.-F., Legallois, D., Champ-Rigot, L., Milliez, P., Funck-Brentano, C., & Dolladille, C. (2021). Identification of anticancer drugs associated with atrial fibrillation: analysis of the WHO pharmacovigilance database. *European Heart Journal. Cardiovascular Pharmacotherapy*, *7*(4), 312–320. https://doi.org/10.1093/ehjcvp/pvaa037

Apache Software Foundation. (2026a). *Apache arrow*. https://arrow.apache.org/

Apache Software Foundation. (2026b). *Apache Parquet*. https://parquet.apache.org/

Chretien, B., Dolladille, C., Nishida, K., Aleksic, B., Alexandre, J., & L'orphelin, J.-M. (2025). Analysis of anticancer drug associated adverse reactions in depressive patients from vigibase. *Scientific Reports*, *15*(1), 45751. https://doi.org/10.1038/s41598-025-28563-9

Dolladille, C., & Chrétien, B. (2025). *vigicaen: 'VigiBase' Pharmacovigilance Database Toolbox*. https://github.com/pharmacologie-caen/vigicaen

Dolladille, C., Ederhy, S., Sassier, M., Cautela, J., Thuny, F., Cohen, A. A., Fedrizzi, S., Chrétien, B., Da-Silva, A., Plane, A.-F., Legallois, D., Milliez, P. U., Lelong-Boulouard, V., & Alexandre, J. (2020). Immune Checkpoint Inhibitor Rechallenge After Immune-Related Adverse Events in Patients With Cancer. *JAMA Oncology*. https://doi.org/10.1001/jamaoncol.2020.0726

Embry, I. (2016). *vaersNDvax: Non-Domestic Vaccine Adverse Event Reporting System (VAERS) Vaccine Data for Present*. https://cran.r-project.org/web/packages/vaersNDvax/index.html

Embry, I. (2018). *Vaersvax: US vaccine adverse event reporting system (VAERS) vaccine data for present*. https://doi.org/10.32614/CRAN.package.vaersvax

Legallois, D., Da Silva, A., Alexandre, J., Milliez, P., Sabatier, R., Blanchart, K., Plane, A.-F., Font, J., Chrétien, B., & Dolladille, C. (2025). Identification of anticancer drugs associated to cancer therapy-related cardiac dysfunction: a VigiBase® disproportionality analysis. *European Heart Journal. Cardiovascular Pharmacotherapy*, *11*(5), 459–468. https://doi.org/10.1093/ehjcvp/pvaf027

Minoc, E.-M., Villain, C., Chrétien, B., Benbrika, S., Heraudeau, M., Lafont, C., Béchade, C., Lobbedez, T., Lelong-Boulouard, V., & Dolladille, C. (2025). Association between antidepressant drugs and falls in older adults: A mediation analysis in the World Health Organization's pharmacovigilance database. *Therapie*. https://doi.org/10.1016/j.therap.2025.01.004

Montastruc, J.-L., Sommet, A., Bagheri, H., & Lapeyre-Mestre, M. (2011). Benefits and strengths of the disproportionality analysis for identification of adverse drug reactions in a pharmacovigilance database. *British Journal of Clinical Pharmacology*, *72*(6), 905–908. https://doi.org/10.1111/j.1365-2125.2011.04037.x

Mukhopadhyay, R., & Markatou, M. (2026). SurVigilance: An application for accessing global pharmacovigilance data. *SoftwareX*, *34*, 102546. https://doi.org/10.1016/j.softx.2026.102546

Nishida, K., Chrétien, B., Dolladille, C., Ebina, T., Aleksic, B., Cabé, N., Savey, V., Onoue, T., & Yatsuya, H. (2025). Psychiatric and psychological adverse effects associated with dulaglutide, semaglutide, and liraglutide: A vigibase study. *Clinical Nutrition*, *51*, 252–265. https://doi.org/10.1016/j.clnu.2025.06.011

Norén, G. N., Hopstadius, J., & Bate, A. (2013). Shrinkage observed-to-expected ratios for robust and transparent large-scale pattern discovery. *Statistical Methods in Medical Research*, *22*(1), 57–69. https://doi.org/10.1177/0962280211403604

Richardson, N., Cook, I., Crane, N., Dunnington, D., François, R., Keane, J., Mecum, B., Moldovan-Grünfeld, D., Ooms, J., Wujciak-Jens, J., & Apache Arrow. (2026). *Arrow: Integration to 'apache' 'arrow'*. https://doi.org/10.32614/CRAN.package.arrow

Uppsala Monitoring Centre. (2026). *About VigiBase*. https://who-umc.org/vigibase/

Wickham, H., & Bryan, J. (2023). *R Packages: Organize, Test, Document, and Share Your Code* (2e ed.). O'Reilly Media. ISBN: 978-1098134464

Wickham, H., Çetinkaya-Rundel, M., & Grolemund, G. (2023). *Arrow – r for data science*. https://r4ds.hadley.nz/arrow.html

Wickham, H., & RStudio. (2023). *Tidyverse: Easily install and load the 'tidyverse'*. https://cran.r-project.org/web/packages/tidyverse/index.html